

Web Technology



Hypertext Markup Language (HTML)

Introduction – Web Technology

- **Client-side Programming**
 - **HTML**
 - **CSS**
 - **Javascript**
- **Server-side Programming**
- **Web Services**

HTML5, CSS3, and Javascript

- **HTML5**
- **CSS3**
- **Javascript**
- **Browsers**

HTML5, CSS3, and Javascript

- **HTML5 - Markup Language**
 - **Content and Structure**
- **CSS3**
- **Javascript**
- **Browsers**

HTML5, CSS3, and Javascript

- **HTML5 - Markup Language**
 - **Content and Structure**
- **CSS3 – Presentation**
 - **Advantage – Swap Styles**
- **Javascript**
- **Browsers**

HTML5, CSS3, and Javascript

- **HTML5 - Markup Language**
 - **Content and Structure**
- **CSS3 – Presentation**
 - **Advantage – Swap Styles**
- **Javascript – Dynamic web pages**
 - **Events**
- **Browsers**

HTML5, CSS3, and Javascript

- **HTML5 - Markup Language**
 - **Content and Structure**
- **CSS3 – Presentation**
 - **Advantage – Swap Styles**
- **Javascript – Dynamic web pages**
 - **Events**
- **Browsers**
 - **Chrome, Firefox, Safari, Opera, Microsoft Internet Explorer 9/ Edge**

HTML

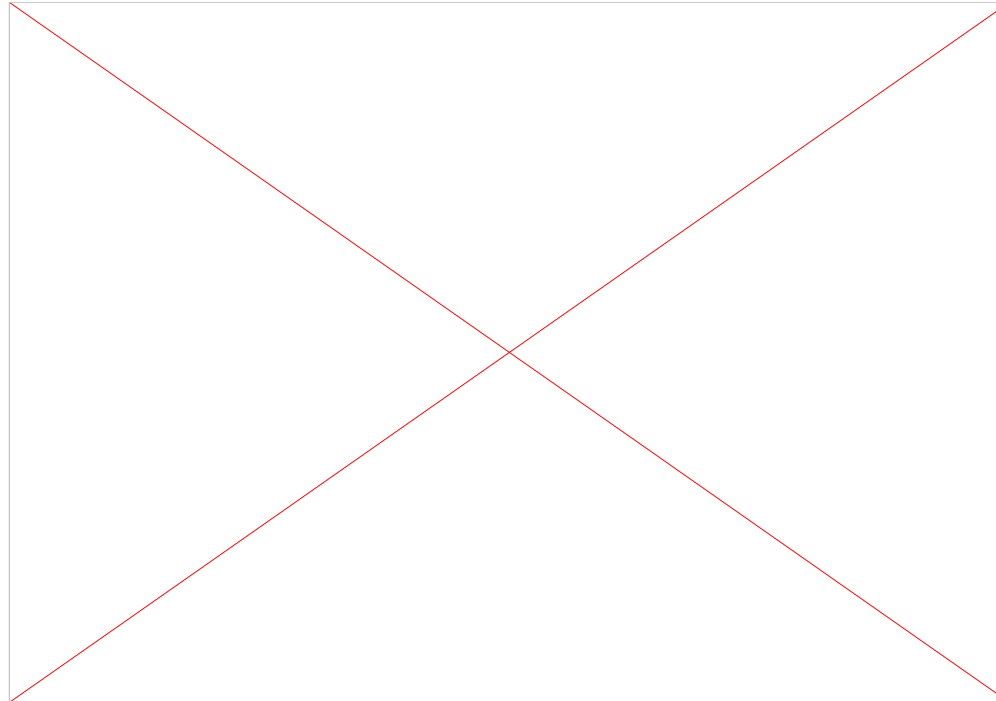
- **HTML**
 - **Hypertext Markup Language**

- **HTML**
 - **Hypertext Markup Language**
 - **Hypertext**
 - **Markup**
 - **Language**

- **HTML**
 - **Hypertext Markup Language**
 - **Hypertext – Text + Highlighted Links**
 - **Easy Navigation within or Across Webpages**



- **HTML**
 - **Hypertext Markup Language**
 - **Hypertext – Text + Highlighted Links**
 - **Easy Navigation within or Across Webpages**



HTML

- **HTML**
 - **Hypertext Markup Language**
 - **Markup**

- **HTML**
 - **Hypertext Markup Language**
 - **Markup**
 - **Mark Text (Highlight, Bold, Italic, Underline) - English Meaning**

- **HTML**
 - **Hypertext Markup Language**
 - **Markup**
 - **Mark Text (Highlight, Bold, Italic, Underline) - English Meaning**
 - **Mark Up Tags**

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

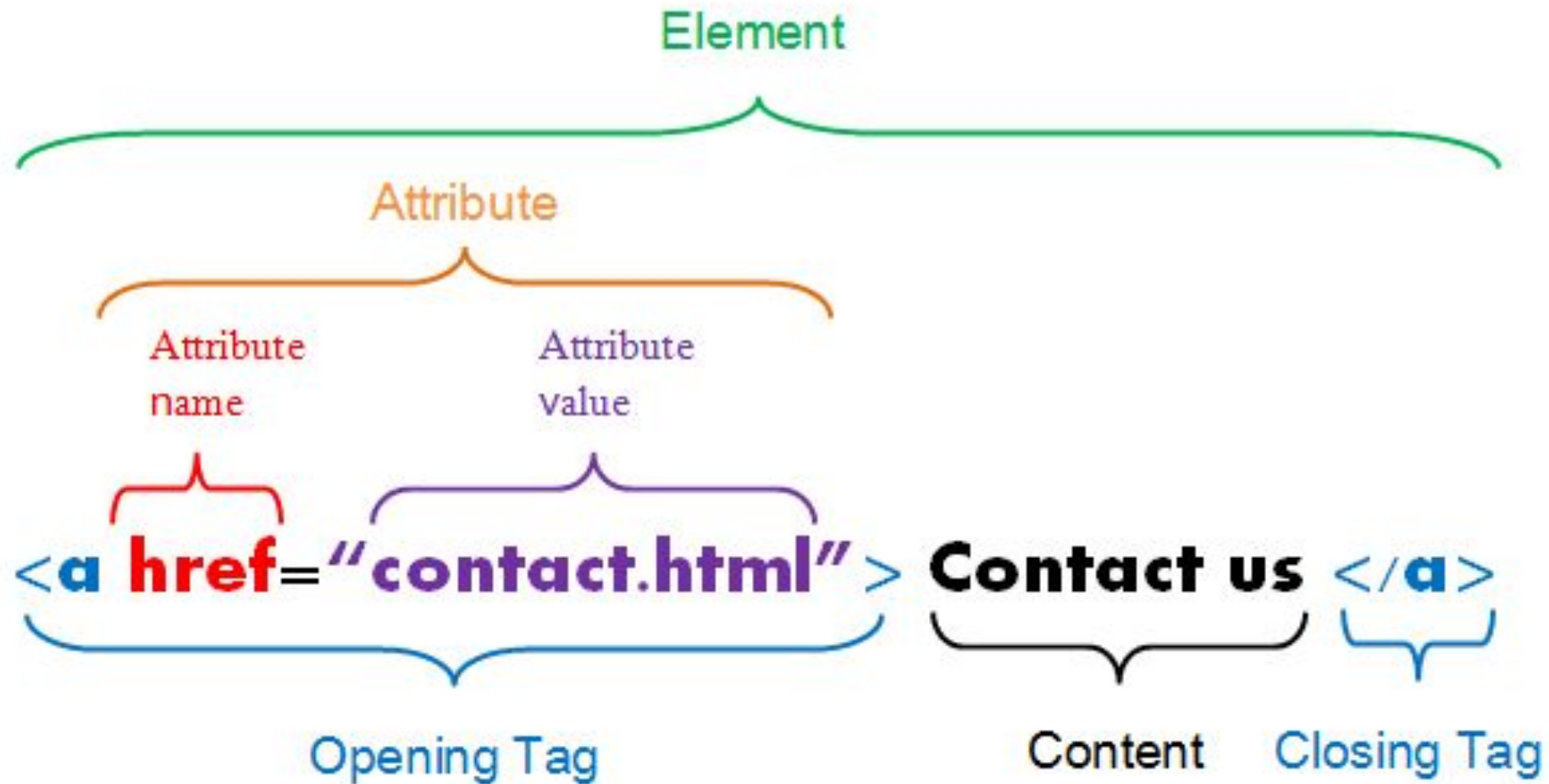
<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

- **Language**

	English	C	HTML
Alphabet	A-Z, a-z	A-Z, a-z	A-Z, a-z
Numbers	0-9	0-9, 0 and 1, 0-7, 0-F	0-9
Words	Words	Tokens	Nodes
Sentences	Grammar + Words = Sentences	Syntax + Tokens = Statements	Syntax + Nodes = Elements
Paragraph	Paragraph	Block	HTMLCollecti on
Chapter/Book	Chapter/Book	Program	Web Page
Library	Library	Library	Website

HTML - Element



HTML - Tags

Heading Tags	<code><h1></code> , <code><h2></code> , <code><h3></code> , <code><h4></code> , <code><h5></code> , and <code><h6></code>
Paragraph Tag	<code><p></code>
Line Break Tag	<code>
</code> or <code>
</code>
Center Tag	<code><center></code>
Horizontal Rule Tag	<code><hr></code> or <code><hr /></code>
Preserve Formatting Tag	<code><pre></code>
Non-breaking Space	<code>&nbsp;</code>
Listing Tags	<code></code> and <code></code> • Unordered Listing: <code></code> • Ordered Listing: <code></code> • List Item: <code></code>
HTML Basic Tags	

HTML - Tags

- **Keywords enclosed in angular brackets (<>)**
- **Total Tags in HTML 5.2 - 142 tags**
- **115 Tags are compactible across HTML versions**
- **Forgetting to close tags – Syntax error**
- **Tags are not case sensitive**
- **Most tags have <tag_name> content <\tag_name>**
- **Best Practice 1 – Use HTML tags with the same case (preferably lower case)**

HTML – Tag Attributes

- **Keywords present in Tags**
- **Specific set of values**
- **Additional Characteristics to Elements**
- **Eg: height, width for an image element**
- **Not mandatory for most tags**

Best Practices

1. **Use HTML tags with the same case (preferably lower case)**

Summary

- **Overview – Client-side Programming**
- **HTML, CSS, Javascript, Browsers**
- **Hypertext**
- **Markup**
- **Language**
- **HTML Elements**
- **HTML Tags**
- **HTML Attributes**
- **Best Practices**

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, 4th Edition, 2008**

Slide Credit: Subin Sahayam, Assistant Professor,

**Department of Computer Science and Engineering
Shiv Nadar University**

THANK YOU

